MATH O MAGIC

5·4

by

GREG HURN

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INTRODUCTION

MATH O MAGIC 5·4 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

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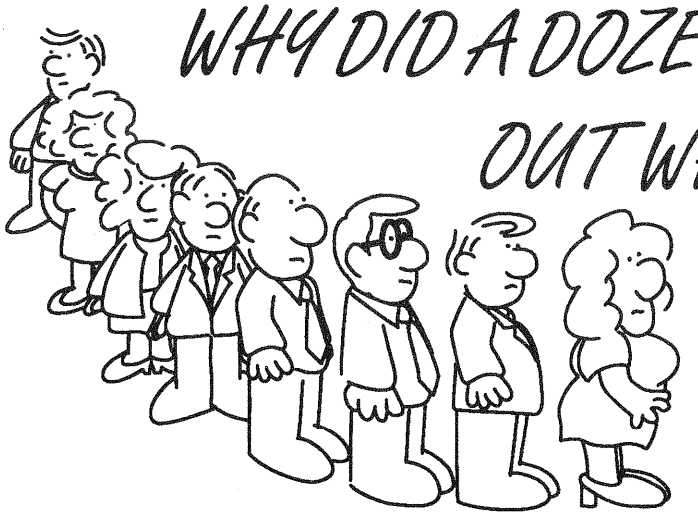
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WHY DID A DOZEN PEOPLE FALL OUT WHEN THE MAN OPENED THE PIECE OF FURNITURE?



Find the difference between the minimum temperatures at each of the following cities.

I	Perth & Tokyo
A	Hong Kong & Moscow
T	Hobart & Helsinki
R	Los Angeles & Banff
W	Brisbane & Vancouver
N	Manila & Helsinki
O	Banff & Hobart
L	Glasgow & Brisbane
U	Calgary & Manila
G	Los Angeles & Helsinki
E	Moscow & Banff
S	Hobart & Tokyo
E	Glasgow & Moscow
P	Glasgow & Banff
A	Brisbane & Helsinki
B	Vancouver & Moscow
S	Perth & Calgary
M	Calgary & Moscow

MANILA	27°
VANCOUVER	−13°
PERTH	23°
HOBART	6°
CALGARY	−42°
HELSINKI	−20°
TOKYO	−12°
HONG KONG	−5°
LOS ANGELES	−1°
BRISBANE	17°
GLASGOW	−7°
MOSCOW	−28°
BANFF	−35°

35°	26°		30°	23°	18°		23°		14°	35°	65°	18°	35°	47°	19°
28°	7°	41°	28°	24°	7°	65°		15°	69°	34°	21°	37°	69°		

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WHO'S THERE?
YELP!
YELP WHO?

Simplify each of the following expressions.



E $|4|$

N $|-7|$

S $|-2+3|$

T $|-3|+|2|$

L $|-7|-|9|$

H $|-7-9|$

N $-|3-2|$

K $|-3| \times |2|$

E $-|-3| \times |-4|$

O $-|5-10|$

I $-|-2{\cdot}5-4{\cdot}5|$

A $|-4 \times 2{\cdot}5|$

M $|3|-|-9|$

U $|-3{\cdot}5|+|4{\cdot}5|$

P $|-4{\cdot}25| \times |4|$

C $|7{\cdot}2|+|-1{\cdot}8|$

S $-|-6{\cdot}9|-|-2{\cdot}1|$

Y $-|-5 \times -3{\cdot}2|$

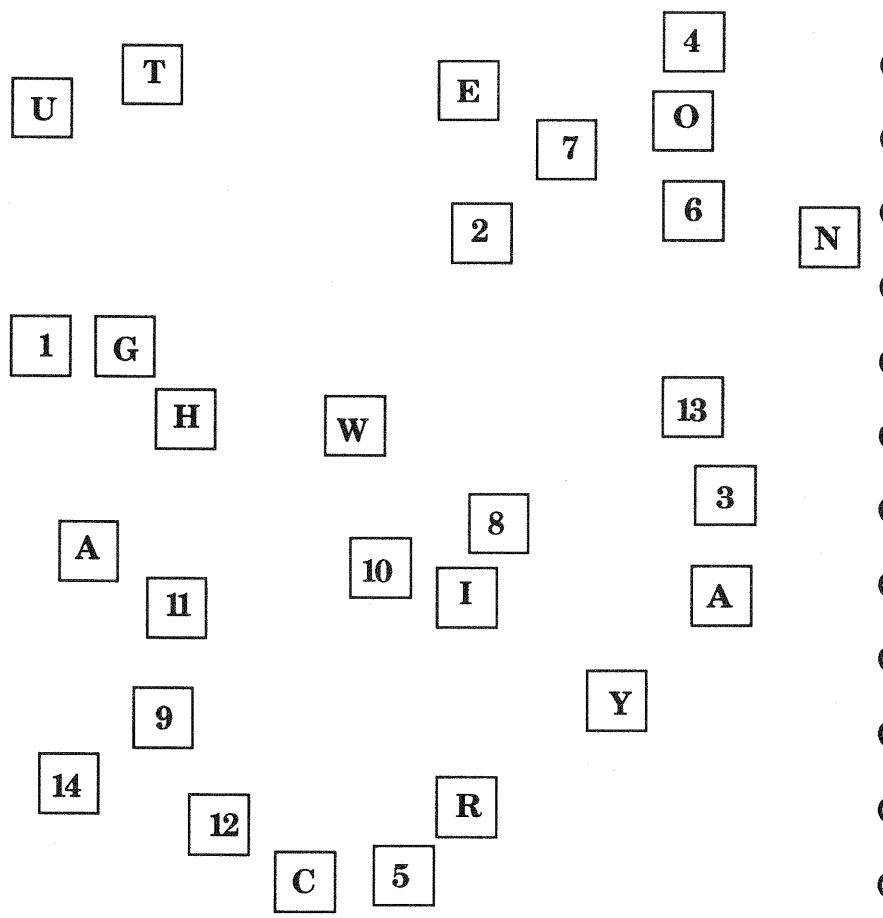
−16	−12	−2	17	−6	4	—	−6	−16	7	−5	1	−12	−7	−9
−9	5	8	9	6	−7	−1	10	6	4	−16	16	−5	−2	−12

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What Do You Call a Gorilla With Cotton In His Ears?

Calculate the maximum % error for each of the following measurements. Draw a straight line to the answer. The line will pass through 2 squares which will tell you where to put a letter in the sentence below.

ESTIMATE		MAX % ERROR
7·3 cm	Θ	Θ 2·78
9·7 cm	Θ	Θ 6·25
2·5 cm	Θ	Θ 4·17
1·8 cm	Θ	Θ 2·38
0·8 cm	Θ	Θ 0·53
94 m	Θ	Θ 7·14
57 cm	Θ	Θ 0·68
0·12 m	Θ	Θ 12·5
7 m	Θ	Θ 0·52
3 cm	Θ	Θ 0·09
21 m	Θ	Θ 0·88
56·3 m	Θ	Θ 2
0·085	Θ	Θ 0·59
4 m	Θ	Θ 16·67



13 11 14 2 6 10 11 1 14 7 3 8 12 11 2

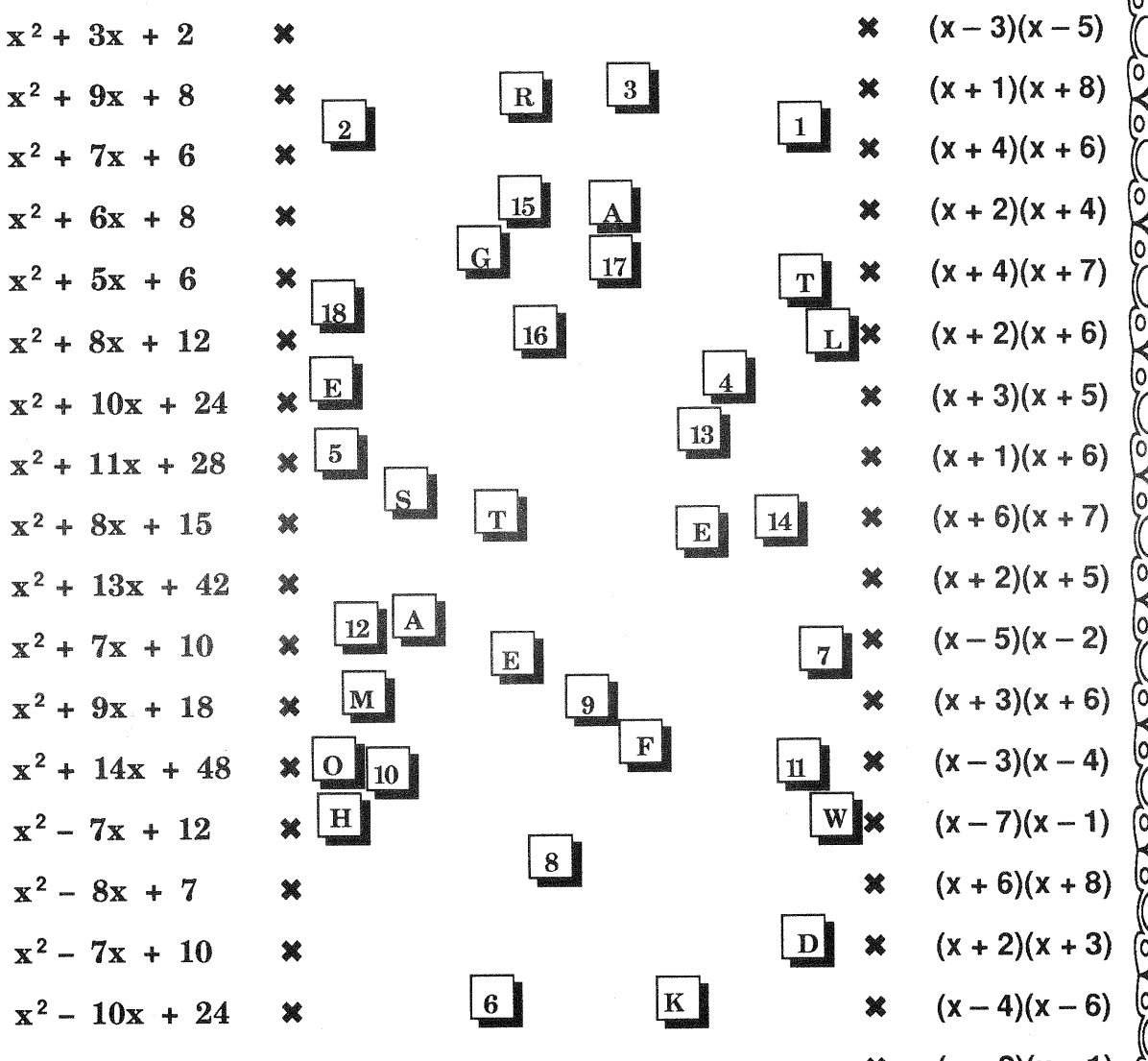
					'									

6 4 5 12 11 2 6 4 13 9 14 7 3

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WHAT USE IS REINDEER?

Factorize each of the following trinomials and join it to the answer in the other column.

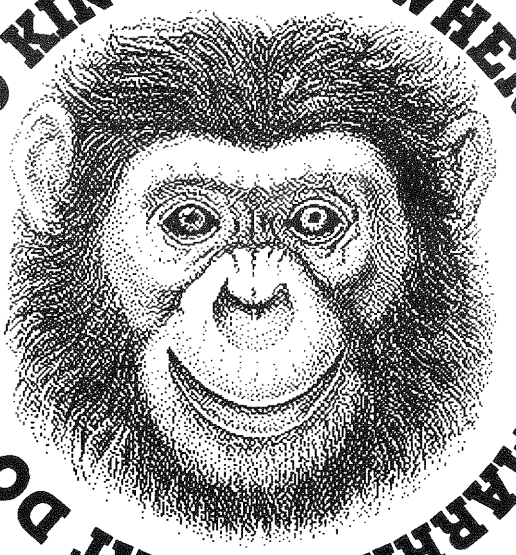


5	10	9	15	6	1	18	11	17	7	16	10	8	1	3	13

4	3	10	8,	2	14	12	3

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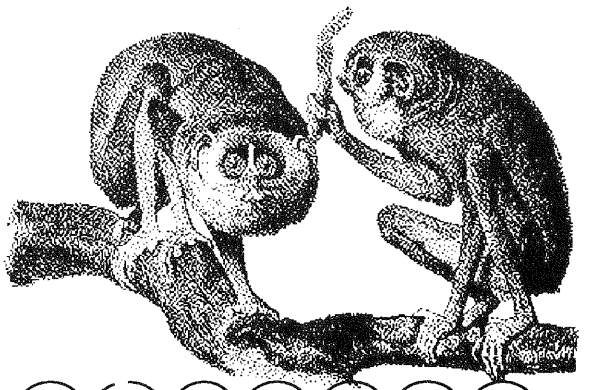
WHAT DO YOU SAY TO KING KONG WHEN HE GETS MARRIED?



Factorize each of the following trinomials and match it with the correct answer.

1. $a^2 - 2a - 3$
2. $a^2 - 9a + 8$
3. $a^2 - 7a + 10$
4. $a^2 - a - 6$
5. $a^2 + a - 12$
6. $a^2 - 8a + 7$
7. $a^2 - 6a + 9$
8. $a^2 - 3a - 10$
9. $a^2 - 4a - 12$
10. $a^2 + a - 20$
11. $a^2 + 2a - 15$
12. $a^2 - 10a + 21$
13. $a^2 - 5a - 14$
14. $a^2 + 5a - 24$
15. $a^2 - 3a - 18$

O	$(a + 4)(a - 3)$
T	$(a + 1)(a - 3)$
R	$(a - 3)(a - 7)$
I	$(a - 7)(a - 1)$
L	$(a + 5)(a - 3)$
G	$(a + 2)(a - 3)$
U	$(a - 5)(a - 2)$
O	$(a + 5)(a - 4)$
T	$(a - 3)(a - 3)$
K	$(a - 6)(a + 3)$
S	$(a - 6)(a + 2)$
A	$(a - 8)(a - 1)$
N	$(a - 5)(a + 2)$
A	$(a + 2)(a - 7)$
N	$(a + 8)(a - 3)$



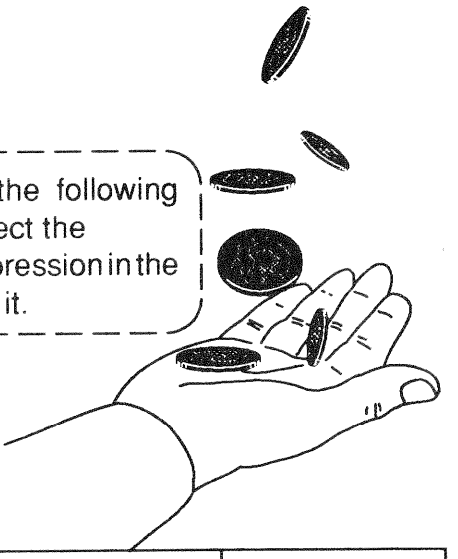


15 5 14 4 12 13 1 3 11 2 7 6 10 8 9

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HOW CAN YOU TELL GERMAN MONEY?

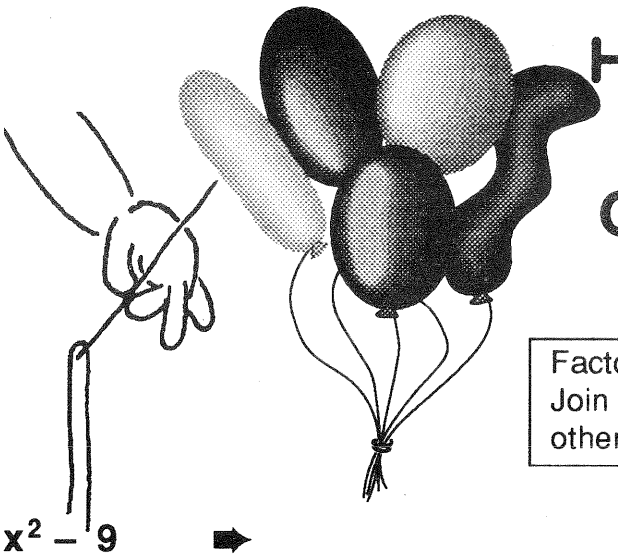
For each of the following trinomials select the equivalent expression in the boxes next to it.



$2x^2 + 5x - 3$	$(2x + 3)(x - 1)$ **S – 2**	$(2x - 1)(x + 3)$ **R – 8**	$(2x + 1)(x - 3)$ **A – 4**	$(2x - 3)(x + 1)$ **B – 4**
$3x^2 - 10x - 8$	$(3x - 2)(x + 4)$ **C – 7**	$(3x + 8)(x - 1)$ **M – 2**	$(3x + 2)(x - 4)$ **B – 1**	$(3x - 4)(x + 2)$ **T – 10**
$5a^2 - 6a - 11$	$(5a - 11)(a + 1)$ **S – 10**	$(5a + 1)(a - 11)$ **O – 9**	$(5a - 1)(a - 11)$ **D – 6**	$(5a + 11)(a - 1)$ **E – 5**
$2a^2 + a - 6$	$(2a + 1)(a - 6)$ **P – 1**	$(2a + 3)(a - 2)$ **A – 3**	$(2a - 3)(a + 2)$ **A – 7**	$(a + 3)(2a - 2)$ **Y – 9**
$3n^2 - 5n - 12$	$(3n - 4)(n + 3)$ **S – 2**	$(3n - 1)(n + 12)$ **D – 10**	$(3n - 3)(n + 4)$ **C – 4**	$(n - 3)(3n + 4)$ **I – 3**
$12c^2 - 5c - 2$	$(6c + 2)(2c - 1)$ **U – 4**	$(6c - 1)(2c + 2)$ **E – 2**	$(4c + 1)(3c - 2)$ **Y – 2**	$(12c - 1)(c + 2)$ **G – 6**
$2y^2 - 9y + 4$	$(2y + 1)(y - 4)$ **T – 8**	$(2y - 1)(y - 4)$ **S – 5**	$(2y - 2)(y - 2)$ **Q – 3**	$(2y + 1)(y - 5)$ **R – 7**
$3u^2 - 10u - 8$	$(3u - 4)(u + 2)$ **Z – 2**	$(3u - 2)(u + 4)$ **S – 8**	$(u - 4)(3u + 2)$ **K – 9**	$(3u + 2)(u + 4)$ **O – 6**
$7w^2 - 10w - 8$	$(7w - 2)(w + 4)$ **B – 4**	$(w - 2)(7w + 4)$ **M – 6**	$(7w - 4)(w + 2)$ **A – 10**	$(3w - 2)(4w + 4)$ **C – 4**
$12p^2 - 32p + 5$	$(6p - 1)(2p - 5)$ **T – 4**	$(4p + 5)(3p - 1)$ **N – 3**	$(12p + 5)(p - 1)$ **Y – 6**	$(2p - 1)(6p - 5)$ **D – 4**

1	2		3	4	5		6	7	8	9	10

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HOW DO YOU KEEP AN ELEPHANT FROM GOING THROUGH THE EYE OF A NEEDLE?

Factorize each of the following expressions. Join the question to the correct answer in the other column with a straight line.

$x^2 - 9$	➡	⬅ $(9 - x)(9 + x)$
$x^2 - 36$	➡	⬅ $(x - 6)(x + 6)$
$x^2 - 100$	➡	⬅ $2(x - 5)(x + 5)$
$1 - x^2$	➡	⬅ $(6 - 5x)(6 + 5x)$
$81 - x^2$	➡	⬅ $(x - 3)(x + 3)$
$2x^2 - 50$	➡	⬅ $(x - 9)(x + 9)$
$1 - 16x^2$	➡	⬅ $(x - 10)(x + 10)$
$25x^2 - 64$	➡	⬅ $7(1 - 3x)(1 + 3x)$
$x^2 - 81$	➡	⬅ $3(x - 5)(x + 5)$
$3x^2 - 75$	➡	⬅ $(1 - x)(1 + x)$
$49x^2 - 1$	➡	⬅ $(7x - 1)(7x + 1)$
$36 - 25x^2$	➡	⬅ $(3 - 11x)(3 + 11x)$
$9 - 121x^2$	➡	⬅ $(1 - 4x)(1 + 4x)$
$18 - 242x^2$	➡	⬅ $2(3 - 11x)(3 + 11x)$
$7 - 63x^2$	➡	⬅ $5(3x - 5)(3x + 5)$
$32x^2 - 18$	➡	⬅ $(5x - 8)(5x + 8)$
$45x^2 - 125$	➡	⬅ $2(4x - 3)(4x + 3)$

4	15	8	1	17	10	2	9	16	7	11	12	6	13	5	14	3

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WHO'S THERE?
ARMENIA.
ARMENIA WHO?

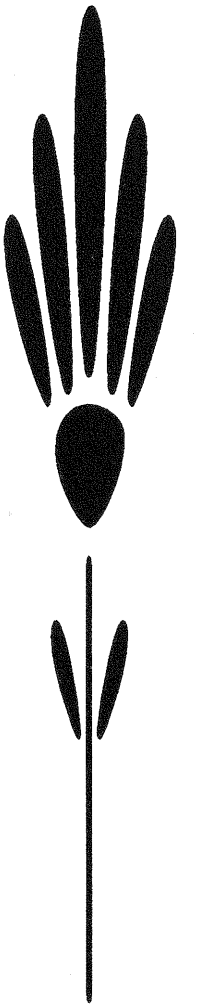


Find the pair of solutions for each of these equations in the code key.

1. $(x - 1)(x + 2) = 0$
2. $(x + 3)(x - 2) = 0$
3. $(x - 4)(x + 4) = 0$
4. $(x - 7)(x + 1) = 0$
5. $(x + 5)(x + 4) = 0$
6. $(x - 9)(x - 1) = 0$
7. $(x + 6)(x - 5) = 0$
8. $(x - 10)(x + 4) = 0$
9. $x(x + 2) = 0$
10. $(x - 3)(x + 3) = 0$
11. $(2x + 1)(x - 2) = 0$
12. $(x + 7)(2x - 1) = 0$
13. $(3x - 2)(x + 6) = 0$
14. $(4x - 5)(2x - 3) = 0$
15. $(3x + 4)(4x - 3) = 0$
16. $(9x - 4)(3x - 4) = 0$
17. $(5x + 2)(3x - 1) = 0$
18. $(2x - 5)(2x + 5) = 0$
19. $x(5x - 3) = 0$
20. $(3x - 7)(2x - 3) = 0$

CODE KEY

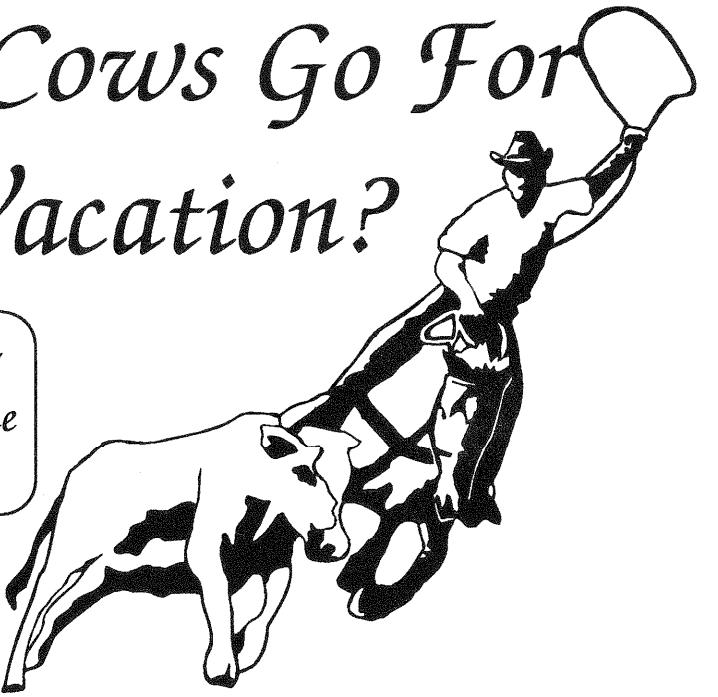
O	$= 5, -6$
M	$= 0, -2$
I	$= \frac{2}{3}, -6$
A	$= 7, -1$
Y	$= \frac{4}{9}, \frac{4}{3}$
A	$= \frac{7}{3}, \frac{3}{2}$
R	$= -3, 3$
N	$= -3, 2$
E	$= -7, \frac{1}{2}$
R	$= \frac{5}{4}, \frac{3}{2}$
E	$= 4, -4$
D	$= \frac{-1}{2}, 2$
V	$= \frac{-4}{3}, \frac{3}{4}$
R	$= 1, -2$
Y	$= 9, 1$
W	$= \frac{-2}{5}, \frac{1}{3}$
E	$= 0, \frac{3}{5}$
A	$= -5, -4$
I	$= 10, -4$
S	$= \frac{-5}{2}, \frac{5}{2}$



20	14	9	19	2	13	4		3	15	12	10	6
17	7	1	11		8		18	5	16			

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Where Do Cows Go For Their Vacation?



Solve each of the following equations by factorizing the expressions firstly.

A $x^2 + 7x + 6 = 0$

I $x^2 - 2x - 8 = 0$

N $x^2 - 5x + 6 = 0$

O $x^2 - 7x + 12 = 0$

T $x^2 + 9x + 20 = 0$

R $x^2 - 2x - 15 = 0$

O $x^2 - 9 = 0$

I $x^2 - 11x + 18 = 0$

F $x^2 + 4x - 21 = 0$

L $x^2 - 3x = 10$

O $x^2 + 6x = 16$

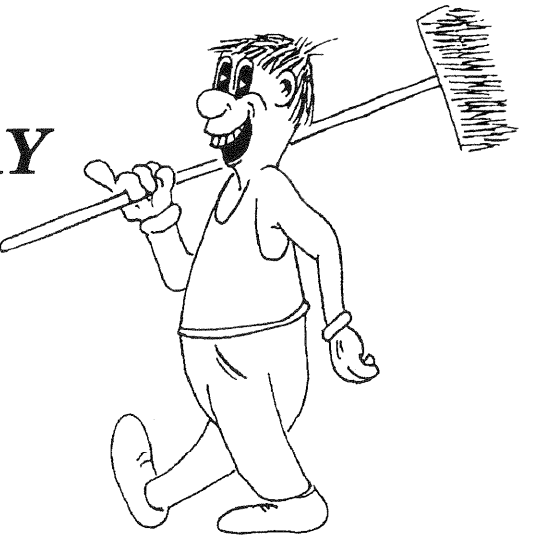
W $x^2 - 9x = -20$

C $x^2 - 11x = -28$

$-4, -5$	$3, 4$		$4, 7$	$3, -3$	$4,5$	$5, -2$	$4, -2$	$-7, 3$	$-8, 2$	$5, -3$	$3, 2$	$2, 9$	$-1, -6$

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WHAT DID THE MOTHER BROOM SAY TO HER INFANT?

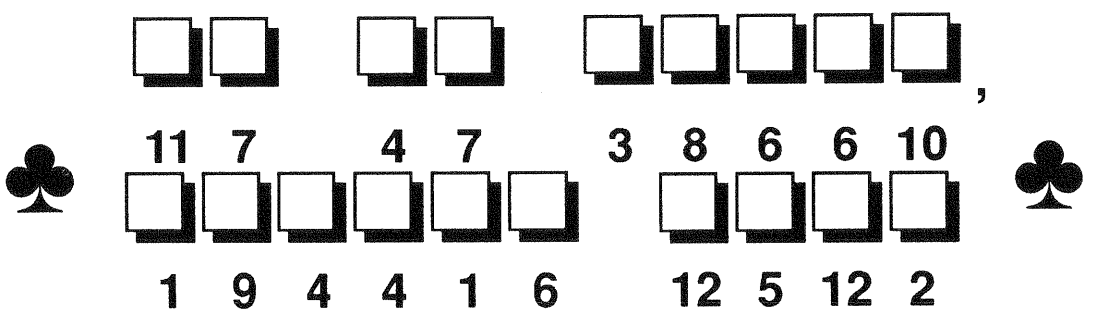


Solve the following equations and find the answers in the code key.

A $x^2 - 16 = 0$
O $6x^2 - x - 1 = 0$
L $2x^2 + x - 6 = 0$
T $5x^2 = 20$
E $4x^2 - 36 = 0$
G $3x^2 - 75 = 0$
Y $12x^2 - 5x = 3$
I $x^2 - 8 = 7x$
B $x(3x - 7) = -2$
S $3x^2 - 7x - 6 = 0$
P $x(x - 1) = 6$
W $4x^2 = 3 - 4x$

CODE KEY

1. $\frac{3}{2}, -2$
2. $\frac{3}{4}, -\frac{1}{3}$
3. $-\frac{2}{3}, 3$
4. $-2, 2$
5. $4, -4$
6. $3, -3$
7. $-\frac{1}{3}, \frac{1}{2}$
8. $-\frac{3}{2}, \frac{1}{2}$
9. $8, -1$
10. $3, -2$
11. $5, -5$
12. $2, \frac{1}{3}$



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WHAT DO YOU CALL A BABY RIFLE?

TRANSLATE EACH OF THE FOLLOWING INTO AN ALGEBRAIC EQUATION. SOLVE THE EQUATION AND USE THE ANSWERS TO FIND THE SOLUTION TO THE RIDDLE.

A – The product of a number and another number 9 larger is 36. Find the positive number.

O – When a positive number is added to its square, the total is 30. Find the number,

N – When two consecutive positive numbers are multiplied together, the result is 56. Find the smaller of the numbers.

S – The sum of the square of a number and five times the number is 36. Find the positive number.

A – The sum of a negative number and its square is 30. Find the number.

F – A number when squared and then doubled, gives the same result as multiplying it by 3 and then adding 5. What is the number (an integer).

U – A positive number squared is equal to the same number multiplied by 6 and added to 40. Find the number.

O – Find the positive number which when subtracted from twice the square of the number gives 6.

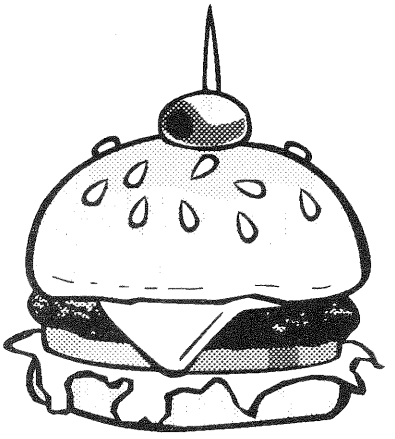
N – Bill is 7 years older than Sharon. The product of twice Bill's age and Sharon's age is 16. How old is Sharon?

G – A rectangle with an area of $80cm^2$ has a width 2cm less than its length. Find its width.



–6		4	2	1		5	–1		3		8	10	7

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HOW DO YOU MAKE A HAMBURGER ROLL?

Find the equation of the axis of symmetry in each of the following equations. Use the answers in the code key to solve the riddle.

1. $y = (x-1)^2$
2. $y = (x+3)^2$
3. $y = (x-5)^2$
4. $y = -(x+2)^2$
5. $y = -(x+5)^2$
6. $y = 2(x+7)^2 - 3$
7. $y = (x-6)^2 + 2$
8. $y = (x-3)^2 - 9{\cdot}3$
9. $y = 2(x+10)^2$
10. $y = -3(x-4)^2$
11. $y = -5(x+1)^2$
12. $y = 4(x-9)^2 + 7$
13. $y = -6(x+8)^2 - 9$
14. $y = 8 + (x-2)^2$
15. $y = 3 - (x-10)^2$
16. $y = 4 - 3(x+9)^2$
17. $y = 7 + 2(x-8)^2$
18. $y = 3{\cdot}8 - 4(x+6)^2$

CODE KEY	
T	$x = 3$
S	$x = 8$
D	$x = 5$
K	$x = -1$
O	$x = -10$
P	$x = 1$
L	$x = -5$
T	$x = -9$
E	$x = 6$
I	$x = 9$
H	$x = -3$
A	$x = 4$
E	$x = -8$
N	$x = -2$
F	$x = 2$
V	$x = -7$
G	$x = 10$
O	$x = -6$

| | | |

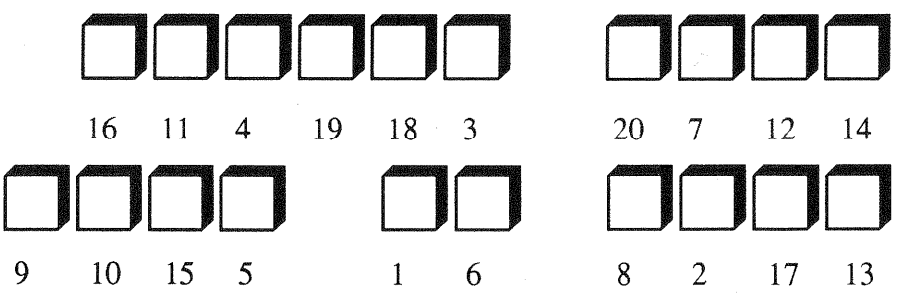
WHO'S THERE?
WOODEN SHOE.
WOODEN SHOE WHO?

Find the equation of the axis of symmetry for each of the following.
Identify the answer in the code key.

O	$y = x^2 + 2x - 3$	**T**	$y = x^2 + 5x - 9$
E	$y = x^2 + 4x + 7$	**W**	$y = x^2 - 3x + 2$
N	$y = x^2 - 2x - 6$	**O**	$y = x^2 - 7x + 8$
D	$y = x^2 - 6x - 5$	**E**	$y = x^2 + 9x + 4$
L	$y = x^2 + 10x - 1$	**I**	$y = x^2 + 7$
E	$y = x^2 - 8x + 4$	**O**	$y = 2x^2 + 12x - 30$
O	$y = x^2 - 4x - 5$	**K**	$y = 3x^2 - 30x + 9$
K	$y = x^2 - 14x + 5$	**N**	$y = 5x^2 - 25x + 20$
O	$y = x^2 + 12x - 9$	**W**	$y = 2x^2 + 28x^2 + 16$
H	$y = x^2 + 8x - 6$	**S**	$y = 2x^2 + 14x - 30$

CODE KEY

1.	$x = -2{\cdot}5$	***6.***	$x = -6$	***11.***	$x = -3$	***16.***	$x = -7$
2.	$x = 2{\cdot}5$	***7.***	$x = -4$	***12.***	$x = -1$	***17.***	$x = 2$
3.	$x = 1$	***8.***	$x = 5$	***13.***	$x = 1{\cdot}5$	***18.***	$x = -4{\cdot}5$
4.	$x = 3{\cdot}5$	***9.***	$x = -5$	***14.***	$x = 4$	***19.***	$x = 3$
5.	$x = -2$	***10.***	$x = 0$	***15.***	$x = 7$	***20.***	$x = -3{\cdot}5$

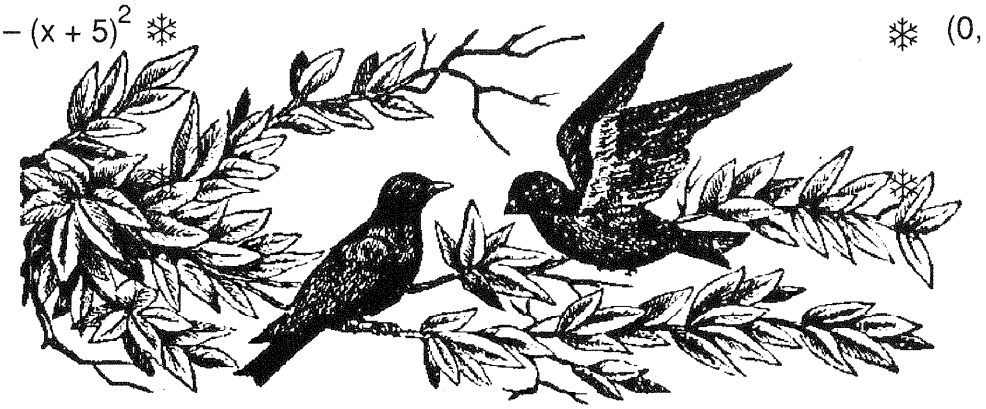


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WHAT DID THE PIGEONS DECIDE TO DO AS THEY FLEW OVER THE CAR LOT?

Join each equation with its turning point in the other column using a straight line. The line will go through one letter and number to help you decode the riddle.

$y = (x-1)^2$	(−3, 0)
$y = (x+3)^2$	(2, 3)
$y = (x-2)^2 + 3$	(0, 7)
$y = (x+7)^2 - 4$	(−2, 1)
$y = x^2$	(−6, −3)
$y = x^2 + 7$	(−7, −4)
$y = x^2 + 5$	(5, 2)
$y = -x^2 + 2$	(0, 2)
$y = -(x+2)^2 + 1$	(4, −2)
$y = -(x+3)^2 + 4$	(0, 0)
$y = (6+x)^2 - 3$	(−3, 4)
$y = (5-x)^2 + 2$	(−3, −5)
$y = -(3+x)^2 - 5$	(−5, −3)
$y = -(4-x)^2 - 2$	(1, 0)
$y = -3-(x+5)^2$	(0, 5)



2	4	15	5	1	7	2	10	6	13	15	10	8	11	12	7	9	10	3	3	14

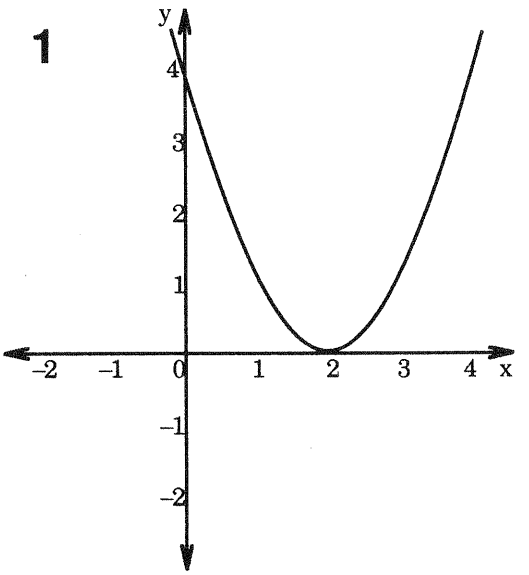
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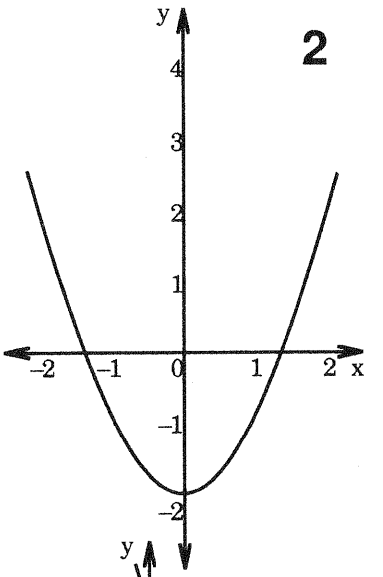
CITY BOY :
"Is it easy to milk a cow?"

Identify each of the graphs from the code key.

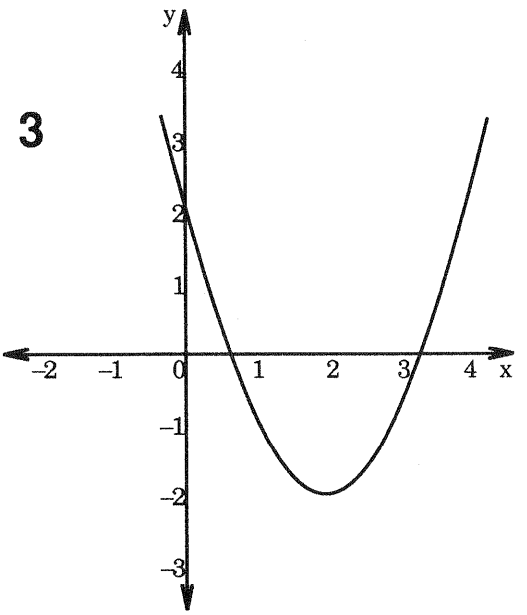
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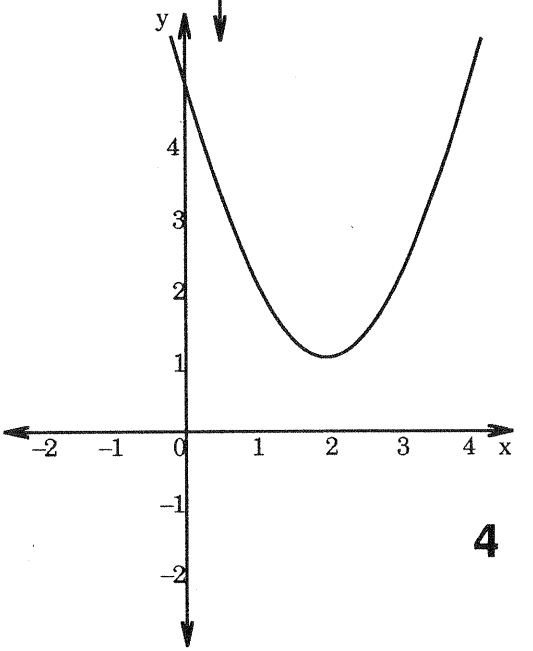
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3



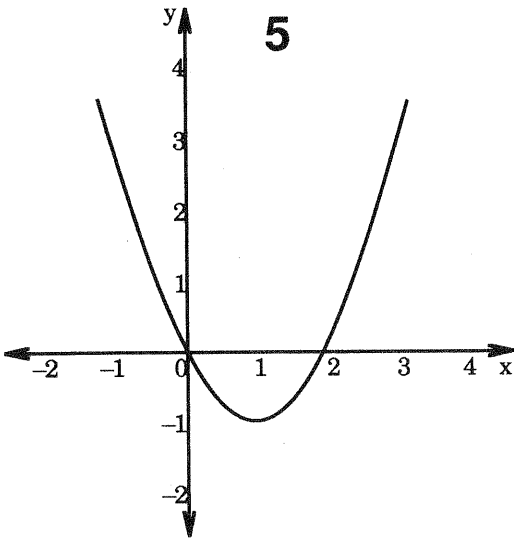
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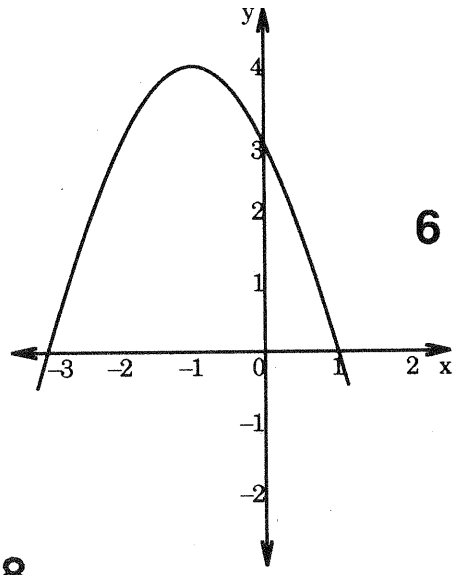
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C	$y = (x-2)^2 - 2$	**CODE**	D	$y = (x+1)^2 - 2$
N	$y = (x-1)^2 - 3$		I	$y = x^2 - 2$
K	$y = -(x-2)^2 + 2$		A	$y = (x+1)^2 - 3$
R	$y = (x-1)^2 - 1$		O	$y = -(x+1)^2 + 4$
A	$y = (x-2)^2$		Y	$y = -(x-1)^2 + 2$
N	$y = (x+2)^2 + 1$		E	$y = (x-2)^2 + 1$
J	$y = -x^2 + 2$		T	$y = (x+1)^2 + 1$

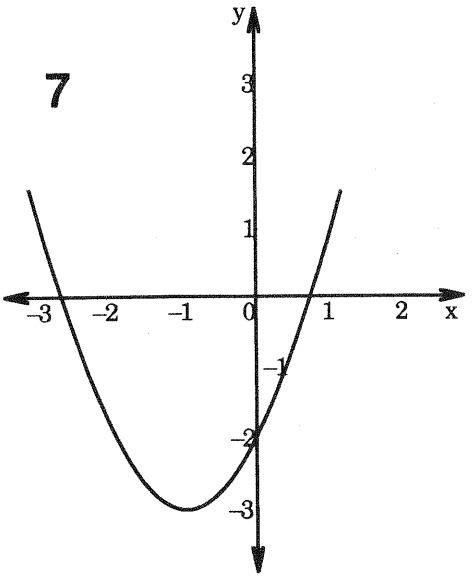
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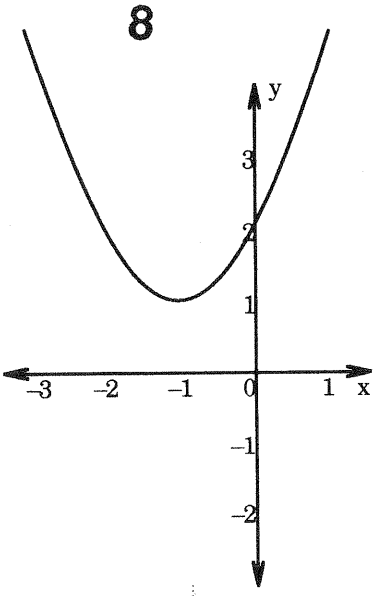
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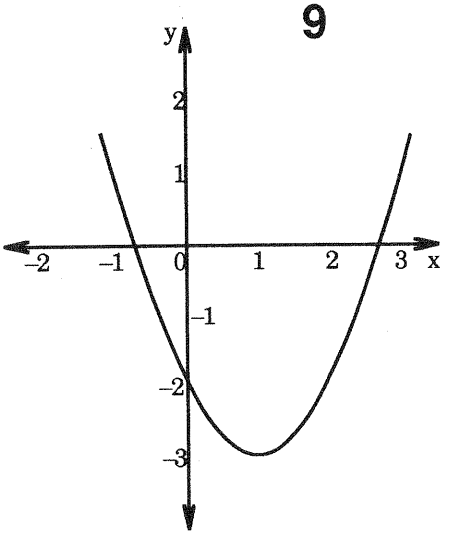
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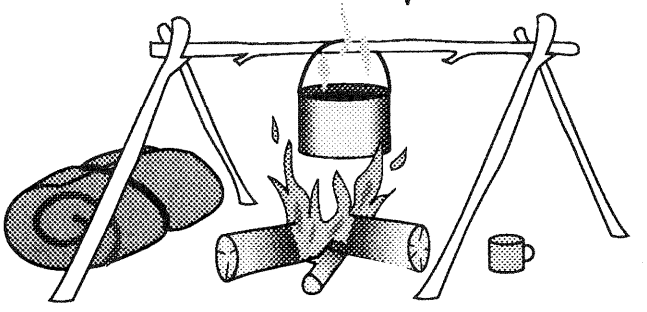


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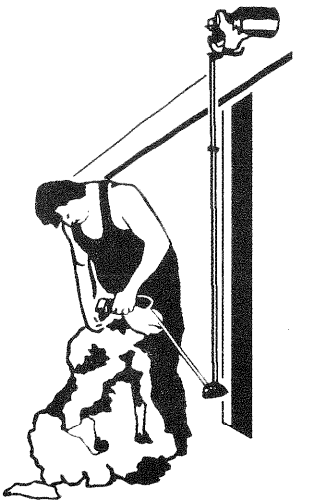
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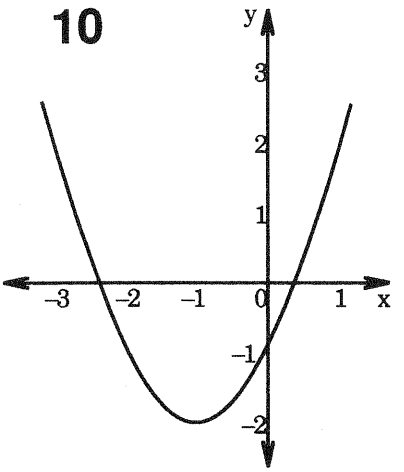


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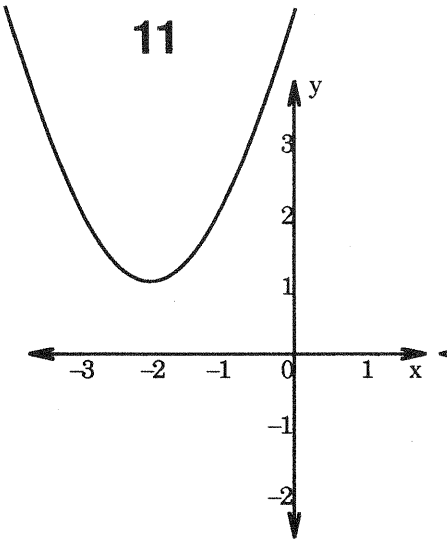




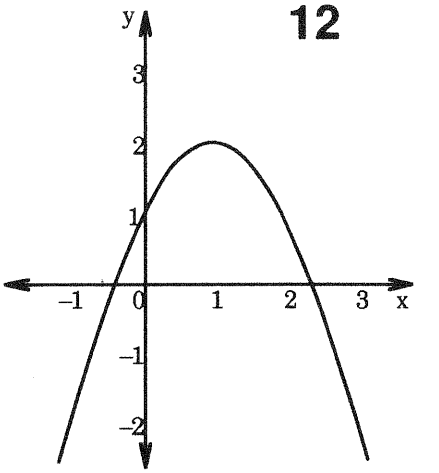
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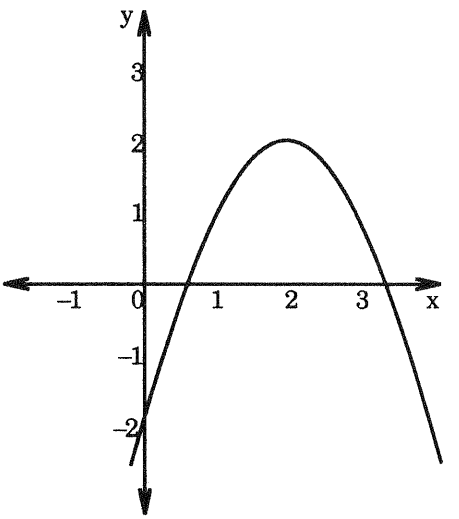
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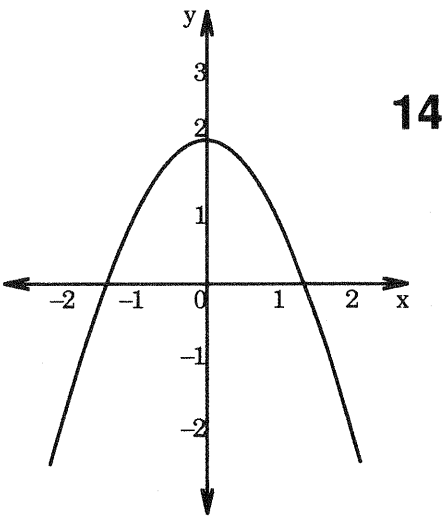
12



13



14



FARM BOY: "

7	9	12		14	4	5	13

"

3	1	11		10	6	2	8

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WHO'S THERE? WILMA. WILMA WHO?

Work out the turning point for each of the following functions.

A $y = 2(x-3)^2 + 1$

O $y = 3x^2 + 1$

L $y = -2x^2 - 5$

M $y = \frac{1}{2}(x+1)^2 - 3$

C $y = 3(x-\frac{1}{2})^2 + 2$

U $y = 2(x+\frac{1}{2})^2 - \frac{1}{2}$

E $y = -\frac{1}{2}(x-2)^2 + 1$

M $y = 6 - x^2$

S $y = 4 + 3x^2$

I $y = 5 - 3(x+2)^2$

T $y = 8 + 2(x-\frac{1}{2})^2$

M $y = -3 - 2(x+4)^2$

E $y = -2(x-4)^2 + 3$

D $y = \frac{1}{2}(x-6)^2 - 4$

E $y = -4 - 2x^2$

A $y = 7 - 3x^2$

R $y = 5 - \frac{1}{2}(x+6)^2$

W $y = -5 + \frac{1}{2}(x+6)^2$

R $y = -5 + 6(x-\frac{1}{2})^2$

$(-6,-5)$	$(-2,5)$	$(0,-5)$	$(-4,-3)$	$(3,1)$		$(6,-4)$	$(-6,5)$	$(2,1)$	$(0,7)$	$(-1,-3)$	$(0,4)$

$(\frac{1}{2},2)$	$(0,1)$	$(0,6)$	$(4,3)$		$(\frac{1}{2},8)$	$(\frac{1}{2},-5)$	$(-\frac{1}{2},-\frac{1}{2})$	$(0,-4)$

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FIRST CANNIBAL:

"Am I late for dinner?"



Join the equivalent terms from each column.
Complete the square on terms in the left column.

$x^2 + 6x + 9$	*	V	3	* $(x + 3)^2$
$x^2 + 4x + 4$	*	T 4		* $(x + 1)^2 + 1$
$x^2 - 8x + 16$	*		E	* $(x - 2)^2 + 3$
$x^2 - 12x + 36$	*	10		* $(x + 2)^2$
$x^2 + 2x + 2$	*		13 E	* $(x - 6)^2$
$x^2 - 4x + 7$	*	12 Y 11		* $(x + 3)^2 - 2$
$x^2 + 6x + 7$	*	9 N	A	* $(x - 4)^2 - 10$
$x^2 + 6x + 3$	*	2 O		* $(x - 4)^2$
$x^2 - 8x + 6$	*		Y	* $(x - 3)^2 - 3$
$x^2 - 6x + 6$	*	1 L		* $(x + 4)^2 - 10$
$x^2 + 10x + 10$	*	5 16		* $(x + 4)^2 - 12$
$x^2 - 14x + 9$	*	14	A N	* $(x - 7)^2 - 40$
$x^2 - 8x - 4$	*	8	D	* $(x - 4)^2 - 20$
$x^2 + 8x + 4$	*	15 S	E	* $(x - 7)^2 - 55$
$x^2 - 14x - 6$	*		R 6	* $(x + 5)^2 - 15$
$x^2 + 8x + 6$	*	7		* $(x + 3)^2 - 6$

SECOND CANNIBAL:

" '

| | | | |



WHO'S THERE?
VERA
VERA WHO?

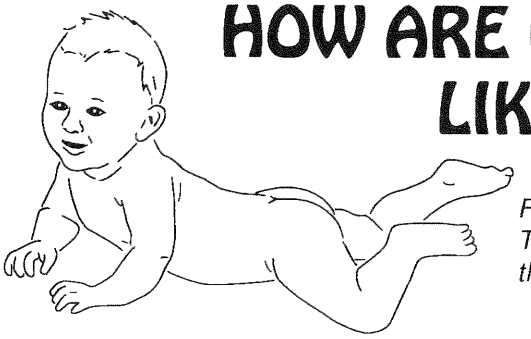
Complete the square for each of the following expressions. Find the answers in the code key and use the code to solve the puzzle.

R	$x^2 + 3x - 5$
T	$x^2 - x + 2$
H	$x^2 - 5x - 3$
W	$x^2 + x - 2$
N	$x^2 + 11x + 4$
S	$x^2 + 7x - 5$
A	$x^2 - 3x + 6$
O	$x^2 - 9x + 20$
G	$x^2 + 13x + 40$
E	$2x^2 + 4x - 1$
L	$2x^2 - 8x - 3$
E	$4x^2 - 8x + 5$
V	$3x^2 - 12x + 4$
F	$3x^2 - 9x + 2$

CODE KEY

1.	$(x + 5{\cdot}5)^2 - 26{\cdot}25$
2.	$(x - 0{\cdot}5)^2 + 1{\cdot}75$
3.	$(x - 4{\cdot}5)^2 + 0{\cdot}25$
4.	$2(x + 1)^2 - 3$
5.	$4(x - 1)^2 + 1$
6.	$(x + 0{\cdot}5)^2 - 2{\cdot}25$
7.	$(x + 6{\cdot}5)^2 - 2{\cdot}25$
8.	$(x + 1{\cdot}5)^2 - 7{\cdot}25$
9.	$3(x - 1{\cdot}5)^2 - 4{\cdot}75$
10.	$(x - 2{\cdot}5)^2 - 9{\cdot}25$
11.	$2(x - 2)^2 - 11$
12.	$(x + 3{\cdot}5)^2 - 17{\cdot}25$
13.	$3(x - 2)^2 - 8$
14.	$(x - 1{\cdot}5)^2 + 3{\cdot}75$

| | | |



HOW ARE BABIES IN A NURSERY LIKE PRICE LISTS?

Find the axis of symmetry for each of the following functions. The answers occur in the code key and can be used to solve the puzzle.

1. $y = x^2 + 2x - 3$
2. $y = x^2 - 4x + 1$
3. $y = x^2 + 6x - 7$
4. $y = x^2 + 10x - 5$
5. $y = x^2 - 2x + 6$
6. $y = x^2 + 14x - 3$
7. $y = x^2 - 3x + 5$
8. $y = x^2 + 7x - 3$
9. $y = x^2 - 9x + 2$
10. $y = 2x^2 + 8x + 7$
11. $y = 3x^2 - 18x - 4$
12. $y = 2x^2 - 24x + 7$
13. $y = 3x^2 - 15x + 12$
14. $y = 2x^2 + 32x - 17$
15. $y = 4x^2 + 48x + 12$

CODE KEY

U	$x = -7$
T	$x = 3$
J	$x = -8$
N	$x = -1$
T	$x = 2{\cdot}5$
G	$x = -2$
H	$x = 1{\cdot}5$
S	$x = -6$
C	$x = -5$
I	$x = -3{\cdot}5$
O	$x = -3$
E	$x = 4{\cdot}5$
W	$x = 6$
B	$x = 2$
A	$x = 1$



	15	6	2	14	9	4	13
11	3	4	7	5	1	10	9
	12	8	13	7	3	6	11
	1	3	13	8	4	9	

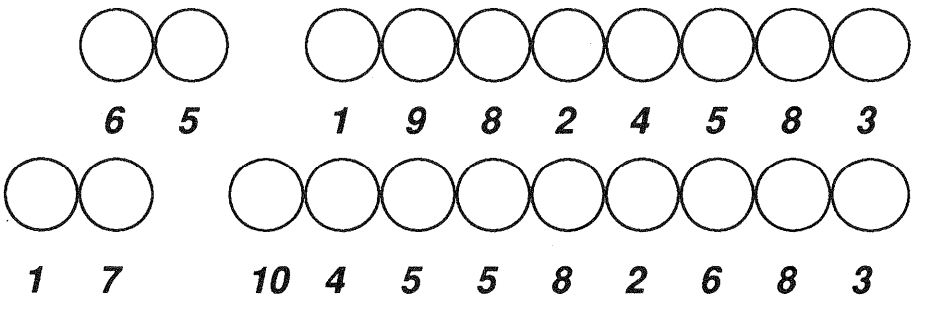
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How is the new doll like a surgeon?

Work out which of the points next to each equation is its turning point. The correct point will tell you where to place the letters in the puzzle below.

$y = x^2 + 3$	(3, 0) **1 – T**	(0, 3) **9 – P**	(0, – 3) **7 – Y**	(1, 4) **3 – U**
$y = 2x^2 - 4$	(2, – 4) **8 – J**	(– 4, 0) **6 – B**	(0, – 4) **2 – R**	(– 4, 2) **2 – G**
$y = x^2 + 2x - 3$	(– 1, – 4) **1 – O**	(0, – 3) **9 – T**	(– 1, – 3) **6 – H**	(– 1, – 6) **4 – S**
$y = x^2 - 4x + 1$	(0, 1) **5 – R**	(1, – 4) **1 – W**	(– 2, 13) **6 – F**	(2, – 3) **6 – I**
$y = x^2 - 8x - 3$	(4, – 19) **7 – N**	(0, – 3) **9 – L**	(1, – 10) **2 – A**	(4, – 3) **8 – M**
$y = x^2 + 6x - 2$	(– 3, – 2) **4 – I**	(0, – 2) **2 – N**	(– 6, – 2) **4 – D**	(– 3, – 11) **3 – S**
$y = x^2 - 10x + 13$	(0, 13) **6 – C**	(1, 4) **7 – C**	(5, – 12) **4 – A**	(– 5, 88) **2 – A**
$y = 2x^2 + 12x - 3$	(0, – 3) **2 – I**	(– 3, – 21) **10 – B**	(2, – 3) **8 – B**	(– 6, – 3) **5 – W**
$y = 3x^2 - 18x + 20$	(3, – 7) **5 – T**	(3, 20) **10 – N**	(3, 0) **9 – E**	(– 2, 20) **3 – L**
$y = 2x^2 - 8x + 12$	(4, 12) **8 – F**	(2, 4) **8 – E**	(2, 12) **8 – A**	(– 2, 36) **8 – D**





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WHAT WOULD BE GOOD ADVICE TO GIVE TO WORMS?

Use the quadratic formula to solve each of the following equations. Find the answers in the sentence below.

A	$2x^2 - 5x - 3 = 0$
E	$2x^2 - x - 3 = 0$
T	$3x^2 - 11x - 4 = 0$
L	$6x^2 + x - 2 = 0$
E	$2x^2 + 5x - 25 = 0$
P	$4x^2 + 4x + 1 = 0$
L	$3x^2 + 11x + 6 = 0$
E	$4x^2 - 12x + 9 = 0$
S	$6x^2 - 17x + 12 = 0$

$\{\frac{4}{3}, \frac{3}{2}\}$	$\{\frac{1}{2}, -\frac{2}{3}\}$	$\{\frac{3}{2}, -1\}$	$\{3, \frac{3}{2}\}$	$\{-\frac{1}{2}, -\frac{1}{2}\}$		$\{-\frac{2}{3}, -3\}$	$\{-\frac{1}{2}, 3\}$	$\{-\frac{1}{3}, 4\}$	$\{-5, \frac{5}{2}\}$

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WHO'S THERE?
TEX.
TEX WHO?

Use the quadratic formula to solve each of the equations.
Draw a straight line to the correct answer in the other column.

$x^2 + x - 6 = 0$	$\{-3, -3\}$
$x^2 + 3x - 4 = 0$	$\{3, -3\}$
$x^2 + 5x + 6 = 0$	$\{3, 5\}$
$x^2 + 2x - 8 = 0$	$\{4, 4\}$
$x^2 + 6x + 9 = 0$	$\{2, -3\}$
$x^2 - 9 = 0$	$\{2, -4\}$
$x^2 - 12x + 36 = 0$	$\{6, 6\}$
$x^2 + 3x - 10 = 0$	$\{2, -5\}$
$x^2 + 7x + 12 = 0$	$\{1, -4\}$
$x^2 - 8x + 15 = 0$	$\{-3, -4\}$
$x^2 - 36 = 0$	$\{2, -7\}$
$x^2 + 5x - 10 = 0$	$\{6, -6\}$
$x^2 - 8x + 16 = 0$	$\{-2, -3\}$

9	3	13	8	12	2	10	5	11	1	6	4	7

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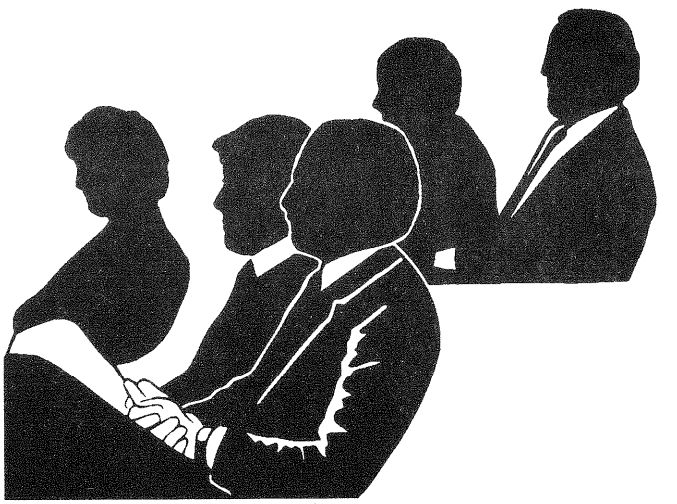
HOW WAS THE TEDIOUS PERSON LIKE A GUN?

Find an exact value for each of the following:

A 17^0
B $2^2 . 9^0$
T 3^{-1}
O 2^{-3}
G $5^2 . 3^0$

L $\left(\frac{1}{3}\right)^{-1}$
E $\left(\frac{2}{3}\right)^{-2}$





A $4^{-1} . 8$
I $3^4 . 9^{-1}$
E $5^{-2} . 4^0$
L 2^{-4}
H 10^{-1}
I $8^{-2} . 4^2$
L $\left(\frac{4}{7}\right)^{-1}$
B $\left(\frac{1}{8}\right)^{-2}$
R $19^0 . 2^5$
A 3^{-3}

W $\left(\frac{1}{10}\right)^{-1}$
S $2^3 . 4^{-2}$
B $2^{-3} . 4^3$

R $2^{-3} . 4^2 . 8$
I $\left(\frac{2}{3}\right)^{-3}$
M $\left(\frac{3}{7}\right)^{-1}$
C $\left(\frac{3}{4}\right)^{-2}$

$\frac{1}{2}$	$\frac{7}{3}$	1	$\frac{7}{4}$	3		$\frac{16}{9}$	2	$\frac{1}{16}$	$\frac{1}{4}$	4	16	$\frac{1}{25}$

10	9	$\frac{1}{3}$	$\frac{1}{10}$		$\frac{1}{27}$		64	$\frac{27}{8}$	25		8	$\frac{1}{8}$	32	$\frac{9}{4}$

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WHAT DID THE DOCTOR SAY WHEN A PATIENT SWALLOWED A ROLL OF FILM?

Find a matching value for each of the following in the code key. Use the code to solve the puzzle.

N	$\sqrt{49}$
P	$\sqrt[3]{64}$
O	$\sqrt{121}$
E	$\sqrt[3]{27}$
L	$81^{\frac{1}{2}}$
V	$196^{\frac{1}{2}}$
U	$125^{\frac{1}{3}}$
L	$\sqrt{169}$
I	$256^{\frac{1}{2}}$
W	$625^{\frac{1}{2}}$
H	$512^{\frac{1}{3}}$
E	$\sqrt{361}$
O	$529^{\frac{1}{2}}$
L	$\sqrt{1444}$
S	$\sqrt[3]{216}$
T	$289^{\frac{1}{2}}$
D	$\sqrt{784}$
G	$\sqrt[3]{1000}$
R	$1728^{\frac{1}{3}}$

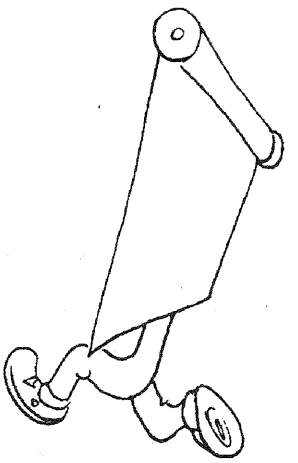
CODE KEY

1. 10
2. 11
3. 16
4. 14
5. 13
6. 7
7. 17
8. 5
9. 23



CODE KEY

10. 28
11. 4
12. 25
13. 38
14. 9
15. 6
16. 3
17. 19
18. 8
19. 12



6	9	7	18	3	6	1		15	16	19	3	2	8	15

12	3	14	13		10	16	4	17	5	2	11

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WHY DID THE DELICATESSEN KEEP BANDAGES IN THE REFRIGERATOR?

Simplify each of the following expressions and find the correct answer in the row next to it. The correct box will tell you where to put the letters in the boxes below.

$a^2 . a^4 . a^{-3}$	$3a$ **1 – A**	$9a^3$ **2 – M**	a^3 **3 – T**	$-3a^2$ **4 – S**
$c^{-3} . c^{-2} . c^6$	c^{11} **10 – Y**	c^0 **11 – T**	$36c$ **2 – D**	c **9 – R**
$m^7 . m^{-4} . m^{-1}$	m^3 **3 – P**	m^2 **8 – O**	m^{28} **5 – M**	$3m$ **6 – T**
$5x^{-4} . 2x^7$	$7x^3$ **7 – Y**	$10x^3$ **10 – F**	$10x^{-28}$ **4 – G**	$7x^{-28}$ **2 – N**
$\frac{c^4}{c^9}$	c^{-5} **6 – O**	$c^{\frac{4}{9}}$ **9 – A**	$\frac{4}{9} c$ **8 – E**	c^5 **7 – C**
$\frac{x^{-3} . x^8}{x^2}$	$\frac{5}{2} x$ **5 – B**	x^{-12} **2 – M**	x^{12} **10 – R**	x^3 **4 – C**
$(n^{-3})^4$	n^1 **6 – L**	n^{-12} **7 – S**	$-12n$ **4 – O**	n^{-7} **2 – E**
$(5a^{-2})^2 . (3h)^2$	$15h^2$ **9 – Z**	$15a^{-4}h^2$ **8 – U**	$45a^0 h^2$ **7 – V**	$225a^{-4}h^2$ **1 – L**
$(3e)^{-2} . 9e^2$	1 **2 – C**	$81e^{-4}$ **6 – E**	27 **10 – T**	$27e^{-4}$ **10 – N**
$(3a^2)^3 . (3a)^{-2}$	$9a^4$ **3 – P**	$9a^3$ **4 – Q**	$9a^{-12}$ **5 – S**	$3a^4$ **11 – U**
$(2a^{-1}b)^3 . (2a^2 b^{-1})^2$	$4a^7 b^5$ **2 – X**	$32ab$ **5 – D**	$4a^{-6}b^{-6}$ **3 – I**	$4a^{-12}b^{-6}$ **4 – J**



10 6 9



4 8 1 5



2 11 3 7

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WHAT DID THE MAN SAY WHEN HE DENTED HIS NEW CAR?

FOR EACH OF THE FOLLOWING PROBLEMS, CALCULATE HOW MUCH **SIMPLE INTEREST** IS PAYABLE.

PRINCIPAL (Amount)	TIME (Years)	RATE % p.a.	SIMPLE INTEREST
$100	3	7	S
$200	5	8	R
$ 80	4	5	N
$250	3	12	Y
$340	2	15	W
$230	4	10	M
$ 78	5	9	A
$ 63	7	6	E
$125	4	14	C
$319	5	12	H
$285	6	8	B
$162	8	5	D
$215	4	15	T
$184	10	9	Z

CODE KEY

1.	$35·10	*6.*	$16	*10.*	$129
2.	$70	*7.*	$90	*11.*	$26·46
3.	$21	*8.*	$191·40	*12.*	$92
4.	$64·80	*9.*	$80	*13.*	$165·60
5.	$102			*14.*	$136·80

10	8	1	10	3		10	8	11		5	1	7		10	8	11			
12	11	9	2	11	4	11	3			14	11	6	13						

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DID YOU HEAR ABOUT THE DOG WHO LIKED TO EAT GARLIC?



For each of the following situations work out the unknown value and use the code key to solve the puzzle.

AMOUNT	TIME (years)	INTEREST RATE (% p.a.)	SIMPLE INTEREST
$600	3	5	**E**
$150	2	**T**	$24
$300	**O**	7	$126
$280	5	6·5	**A**
$450	**N**	8	$252
R	9	5	$36
$65	3	**U**	$23·40
$120	4	**B**	$72
K	6	10	$87
$260	**W**	13	$169
M	4	5	$64
$195	2	**I**	$54·60
$222	15	**H**	$99·90
$88	**C**	15	$118·80
S	5	12	$58·20

Code Key

1. 320	*6.* 12	*11.* 5
2. 80	*7.* 90	*12.* 6
3. 8	*8.* 3	*13.* 7
4. 97	*9.* 15	*14.* 145
5. 14	*10.* 91	*15.* 9

8 5 4 9 10 2 14 5 4 1 6 15 8

11 12 2 4 7 3 8 10 13 8 5 4 9 5 3 7

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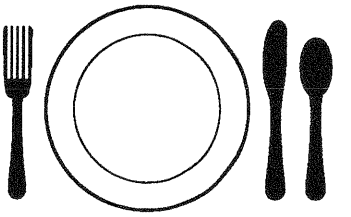


WHERE ON THE HIGHWAY DID THE CANNIBAL LIKE TO STOP?

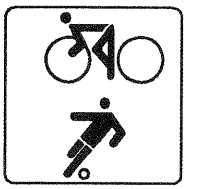
Work out the compound interest payable in each of the following situations and find the answers in the code key.

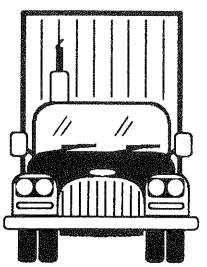






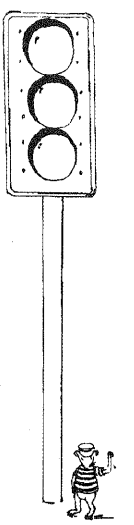
AMOUNT	TIME (years)	RATE (% p.a.)	COMPOUND INTEREST
$100	2	15	*R*
$80	3	12	*V*
$120	4	8	*C*
$150	4	9	*K*
$175	3	14	*T*
$250	5	7	*H*
$110	6	10	*W*
$280	3	4	*U*
$140	5	6	*I*
$70	8	5	*Y*
$330	3	4	*S*
$260	6	7	*D*
$200	5	5	*E*





CODE KEY

1.	$84·27	*6.*	$100·64	*9.*	$61·74
2.	$41·21	*7.*	$32·39	*10.*	$47·35
3.	$55·26	*8.*	$34·96	*11.*	$33·42
4.	$84·87			*12.*	$43·26
5.	$32·25			*13.*	$130·19

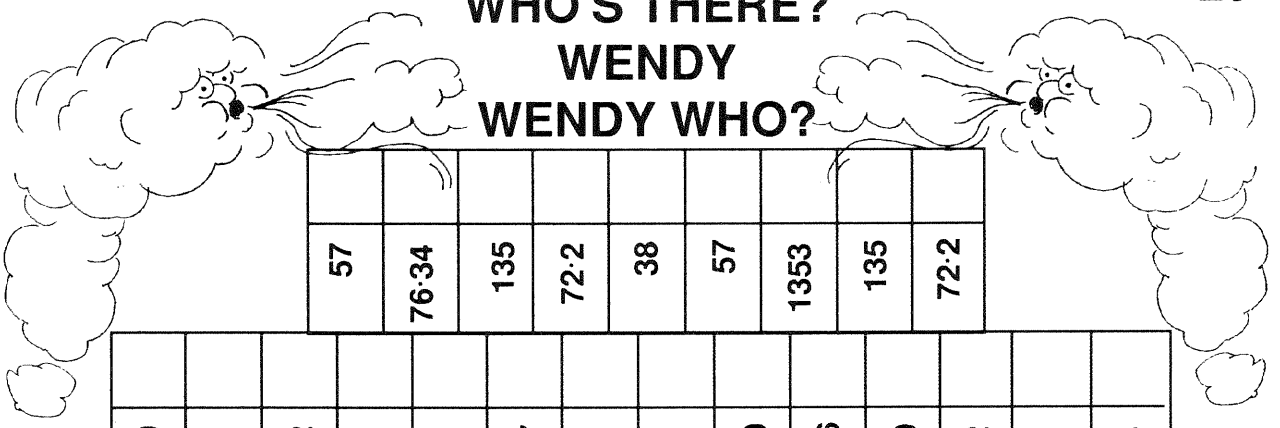


4	6	3	5	3	1	6	3	11	2	3	5	7	3

1	5	8	12	9	13	5	10	7	3	5	2

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WHO'S THERE?
WENDY
WENDY WHO?



57	76·34	135	72·2	38	57	1353	135	72·2

600	26·11	193·2	57	544	59·7	136	76·34	10	21·6	99	72·2	26·11	76·34

57	1353	26·11	26·11	21·6	193·2	10	213

E If Sophie pays $20·82 for 3 records, how much would Adrian pay for 11 records at the same price?

O A stack of 17 books is 142·8 cm high. How high would a pile of 23 of the same books be?

L John exchanged 28 Australian dollars for 17 American dollars. How much would he get for 43 Australian dollars?

T If a rise of 7° in temperature causes the petrol in a tank to increase by 38 m^3, what volume increase would 11° rise cause?

H Flour and sugar are mixed in the ratio of 11 : 2 in a cake mix. For 750 g of flour, how much sugar would be needed?

S A 3 kg pack of chook pellets contains 2175 pellets. How many would be in a 750 g bag?

N A 200 g bag of mixed sweets contains 36 lollies. How many would be in a 750 g bag?

I A water sample of 7 mL contains 861 bacteria. How many bacteria would there be in a similar sample of 11 mL?

C If the exchange rate in a barter system is 26 fowls for 4 pigs, how many pigs would be needed to purchase 65 fowls?

D A trip of 245 km uses 11·8 L of fuel. How much fuel would be needed for a journey of 1500 km?

K If a stack of 375 sheets of paper is 57 mm high, what would be the height of 1400 sheets of similar paper?

A A 2 kg packet of marbles contains 44 marbles. How many marbles would be in the 4·5 kg packet?

R 16 mg of a medicine is required for a person with a weight of 68 kg. How much would be needed for a 92 kg patient?

Y A 375 g bag of table tennis balls contains 25 balls. How many would be in the 570 g bag?

B A 945 cm line of twenty cent coins contains 350 coins. How many coins would be in a line 1620 cm long?

W A hose from a storage tank empties 1975 L in 79 seconds. How long would it take to empty 1425 L?

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MATH O MAGIC 5·4 SOLUTIONS

1. IT WAS A MISSING PEOPLE'S BUREAU
2. YELP ME – MY NOSE IS STUCK IN A KEYHOLE
3. ANYTHING YOU WANT. HE CAN'T HEAR YOU
4. TO MAKE THE FLOWERS GROW, DEAR
5. KONGRATULATIONS
6. BY ITS MARKS
7. TIE A KNOT IN ITS TAIL
8. ARMENIA EVERY WORD I SAY
9. TO COWLIFORNIA
10. GO TO SWEEP, LITTLE BABY
11. A SON – OF – A – GUN
12. TAKE IT TO THE TOP OF A HILL AND GIVE IT A SHOVE
13. WOODEN SHOE LIKE TO KNOW
14. PUT A DEPOSIT ON THE ROLLS
15. ANY JERK CAN DO IT
16. WILMA DREAMS COME TRUE?
17. YES EVERYONE'S ALREADY EATEN
18. VERA ALL THE FLOWERS GONE . . .
19. SUBJECT TO CHANGE WITHOUT NOTICE
20. IT OPERATES ON BATTERIES
21. SLEEP LATE
22. TEX TWO TO TANGO
23. SMALL CALIBRE WITH A BIG BORE
24. NOTHING SERIOUS WILL DEVELOP
25. FOR COLD CUTS
26. THAT'S THE WAY THE MERCEDES BENZ
27. HIS BARK IS MUCH WORSE THAN HIS BITE
28. WHERE THEY SERVE TRUCK DRIVERS
29. WENDY WIND BLOWS THE CRADLE WILL ROCK . . .

